الامتحان النهائي

مقدمة في نظم قواعد البيانات

الأسئلة 33

1 Which SQL statement is correct to find cities names with the condition and temperature table "weather" where the condition is sunny or cloudy and temperature >=60?

أ SELECT city, temperature, condition FROM weather WHERE condition='cloudy' AND condition='sunny' OR temperature >=60

ب SELECT city, temperature, condition FROM weather WHERE condition='cloudy' condition='sunny' AND temperature >=60

ج SELECT city, temperature, condition FROM weather WHERE condition IN ('sunny','cloudy') AND temperature >=60

د SELECT city, temperature, condition FROM weather WHERE condition IN ('sunny','cloudy') OR temperature >=60

2 To change the data type of a column in a table, use the following syntax:

أ ALTER TABLE table_name ADD column_name datatype

ب ALTER TABLE table_name DROP COLUMN column_name

ج ALTER TABLE table_name MODIFY column_name datatype

د ALTER TABLE table_name CHANGE column_name datatype

3 Consider the relation Student(ID, firstName, lastName, address, GPA, CID, StartYear). What is the correct SQL statement to retrieve all students who have a GPA similar to the student whose first name is 'Ahmad'?

أ SELECT * FROM Student WHERE year=(SELECT year FROM Student WHERE firstName='Ahmad');

ب SELECT * FROM Student WHERE GPA=(SELECT GPA FROM Student WHERE firstName='Ahmad');

ج SELECT * FROM Student WHERE GPA=(SELECT GPA FROM Student);

د SELECT * FROM Student WHERE firstName=(SELECT GPA FROM Student WHERE firstName='Ahmad');

4 Consider the relations:Studio(name, address, pressC), Movie(title, year, length, film studioName, producer). Which SQL statement is correct to delete the relation Studio from the database?

أ) ALTER TABLE studio DROP COLUMN studio;
ب) DELETE TABLE Studio;
ج) DROP TABLE Studio;
د) All of the above

5 Consider the relations Student(ID, firstName, lastName, GPA, CID) where ID is the primary key, and Courses(CID, CourseName) where CID is the primary key. What is the correct SQL statement to retrieve students first names and their course names?

أ) SELECT firstName, CourseName FROM Students, Courses;
ب) SELECT firstName, CourseName FROM Students, Courses WHERE Student.CID=Courses.CID;
ج) SELECT firstName, CourseName FROM Students, Courses WHERE CID= CID;
د) SELECT firstName, CourseName FROM Students, Courses WHERE Student. ID=Courses.CID;

6 Which of the following INSERT statements is not valid for the relation R(a INT PRIMARY KEY, b INT DEFAULT 0, c INT NOT NULL)?

أ) INSERT INTO R(a, c) VALUES(7, 8);
ب) INSERT INTO R(c, a, b) VALUES(3, 4, 5);
ج) INSERT INTO R(b, c) VALUES(3, 4);
د) None of the above

7 The HAVING clause does which of the following?

أ) Acts like a WHERE clause but is used for rows rather than columns
ب) Acts EXACTLY like a WHERE clause
ج) Acts like a WHERE clause but is used for groups rather than rows
د) Acts like a WHERE clause but is used for columns rather than groups

8 The SQL WHERE clause:

أ Limits the row data that are returned

ب Limits the column data that are returned

ج Both A and B are correct

د Neither A nor B are correct

9 Which of the following will produce the minimum of all standard prices?

أ SELECT MIN(standard_price) FROM product, v WHERE standard_price=MIN(standard_price);

ب SELECT standard_price FROM product, v WHERE standard_price=MIN;

ج SELECT standard_price FROM product, v ;

د SELECT MIN(standard_price) FROM product, v ;

10 The DELETE TABLE DDL command is used to remove a table from the database.

أ True

ب False

11 Which database language is used to specify database retrievals and updates?

أ DDL

ب DML

ج VDL

د SDL

12 Given "Game" relation which already satisfies 1NF:
Game(GameID, GameDate, GameField, City, GameTitle, Game_session, GameCode). FDs: GameID→GameDate, GameField, GameTitle; GameDate→GameField, GameTitle, Game_session, GameCode. The above relation 'Game' is in 3NF but NOT in BCNF.

أ True

ب False

13 Given "Tech" relation which already satisfies 1NF:Tech(cust_ID, problem_code, tech_no). FDs: cust_ID, problem_code→tech_no; tech_no→proplem_code. The above relation "Tech" is in 3NF but NOT in BCNF.

أ True

ب False

14 Given the following FDs on relation R. which of the following is correct? R(A, B, C, D, E) FDs: A, B, C→D; A, B→C; D→E; A→B

أ (A, B) is a candidate key of R

ب (A, B) is a superkey of R

ج (A) cannot be a key of R

15 Given relation R(A, B, C, D, E, F) where (A, B) is the only candidate key available for R. which of the following FDs can be seen as an example of transitive dependency? FD1:A, B→E, F; FD2:B→F; FD3:E→F

أ FD1

ب FD2

ج FD3

16 Every relation in 2NF is also in 1NF.

أ True

ب False

17 If B→C, then BY→CY, for every Y. this is an example of augmentation in Armstrong's Axioms.

أ True

ب False

18 Given the following relation "Laptop" and functional dependencies: Laptop(L_serial, L_Type, L_year, L_model, L_barcode, L_color) FDs: L_serial, L_Type→ All attributes of R; L_barcode→L_serial. Assume that there is no problem with 1NF, which of the following is correct?

أ The above relation "Laptop" is already in 3NF but Not in BCNF

ب The above relation "Laptop" is already in 2NF but Not in 3NF

ج The above relation "Laptop" satisfies only 1NF

د The above relation "Laptop" satisfies BCNF

19 Given a relation R(A, B, C, D, E, F, G, H, I)which is in 3NF . The primary key is a composite key of two attributes A and B . which of the following functional dependencies must be FALSE?

أ A, B→C, D, E, F, G, H, I
ب D→B
ج A, D→I
د E→A

20 Based on your understanding of the main reasons of the normalization process, select the main informal guidelines for good relational design. A)the BCNF must be satisfied in the database. B) Tuples will have as few NULL values as possible. C) No spurious tuples should be generated. D)Composite keys are always better than single attribute keys because they are clear and easy to understand.

أ C only
ب A and C
ج B and C
د B and D

21 Given the following FDs on relation R. R(A, B, C, D, E, F, G) FDs:A→BC; BC→DE; D→F; CF→G. (B, C) is a candidate key of R.

أ True
ب False

22 If Std_no→std_name, std_major then we can re-write this FD as Std_no→std_name; Std_no→std_major.

أ True
ب False

23 Which of the following statement is NOT correct regarding data normalization:

أ Controls data redundancy
ب Increases data duplication
ج Needs an understanding of functional dependencies of relations attributes

24 Every relation in BCNF is also in 3NF

أ True
ب False

25 Which statement is true regarding the following SQL:Select min(Salary) from Worker where Salary not in (Select min(Salary) from Worker);

أ This will return the lowest salary from table worker

ب This will return the second-lowest salary from table worker

ج This will return the highest salary from table worker

د This will return the salary in table worker

26 If emp_id contains the following set of rows{1,2,2,3,3,1}, what will be the output of the following SQL statement? SELECT emp_id FROM person ORDER BY emp_id DESC;

أ 3,3,2,2,1,1

ب 1,1,2,2,3,3

ج 3,2,1

د 1,2,2,3,3,1

27 Given relation Students(s, Id, LastName, GPA, Birthdate) which of the following is correct to change the data type of column GPA to integer?

أ ALTER TABLE student Modify GPA integer

ب ALTER TABLE student Change GPA integer

ج ALTER TABLE student Drop Column GPA integer

د ALTER TABLE Modify GPA integer

28 The SQL operation that results all pairs of tuples from two relations only where key, foreign key attributes values are matched is called?

أ Join

ب Cartisian product

ج Intersection

د Grouping

29 Given the relation Employee(ID, Name, Age, Department) how do you delete an employee with ID=5 from the Employee table?

أ DELETE FROM Employee WHERE ID=5

ب DELETE * FROM Employee WHERE ID=5

ج REMOVE FROM Employee WHERE ID=5

د DROP TABLE Employee WHERE ID=5

30 Given Emp_Dep(Empno, Ename, DeptNo, Dname, Salary), which functional dependencey types is/are NOT present in the following FDs?(Empno is a key)

أ) Full functional dependency
ب) Partial functional dependency
ج) Transitive functional dependency
د) All of the answers

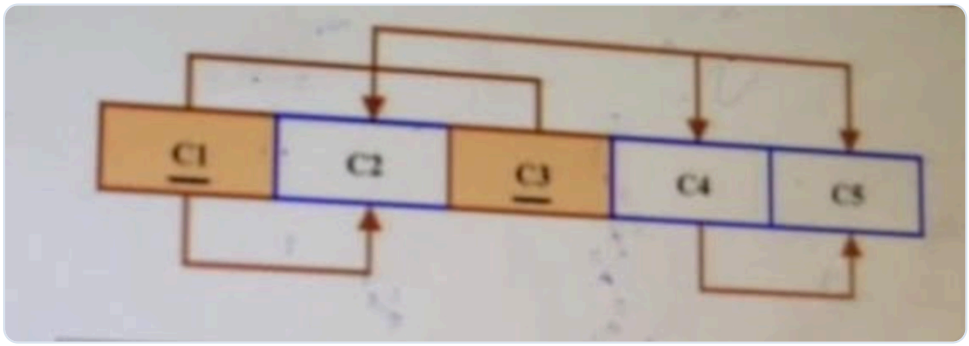
31 Given the schema Course(course_id, sec_id, semester) the course_id, sec_id and semester called.....and course is a.....

أ) Relation, Attribute
ب) Tuple, Relation
ج) Attribute, Relation
د) Tuple, Attribute

32 Given the relation R(A, B, C, D) with BC is the key and the following FDs: A→B/ BC→D/ A→C ,what the normal form is the relation R in?

أ) 1NF
ب) 2NF
ج) 3NF
د) BCNF

33 How many relations will appear after decomposing the following relation to reach 2NF?



أ) 2
ب) 3
ج) This relation is in 2NF
د) 4

مفتاح الإجابات

1 د	2 ج	3 ب	4 ج
5 ب	6 ج	7 ج	8 أ
9 د	10 ب	11 ب	12 أ
13 أ	14 ب	15 ج	16 أ
17 أ	18 أ	19 د	20 ج
21 ب	22 أ	23 ب	24 أ
25 ب	26 أ	27 أ	28 أ
29 أ	30 د	31 ج	32 ج
33 أ			